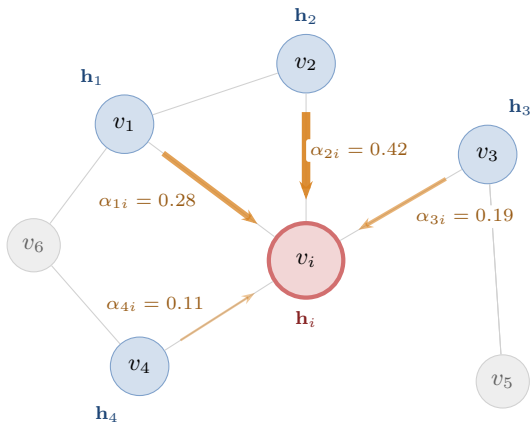




Grosor  $\propto$  peso de atención

→ baja      → alta

**Coefficientes de atención:**

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j])$$

$$\alpha_{ij} = \text{softmax}_j(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}$$

$$\mathbf{h}'_i = \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{h}_j\right)$$

$$\sum_{j \in \mathcal{N}(i)} \alpha_{ji} = 1 \quad (\text{normalizado por softmax})$$